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SITE-SPECIFIC PROPERTY INFORMED INTERFACE DESIGN OF MULTI-MATERIAL LASER POWDER BED FUSION

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ABSTRACT

Laser powder bed fusion (LPBF) is a widely used metal additive manufacturing (AM) method, capable of complex geometry fabrication with high resolution and surface quality. Recent advances have introduced multi-material powder bed AM, which allows for selective powder deposition across multiple layers or within a single layer to truly achieve 3D voxel gradation As this process may combine two dissimilar metals with varying thermal properties along with effects of AM-induced defects, such as as-built cracking, delamination, or residual stresses, are common. Preliminary work has demonstrated that thermomechanical induced cracking at the fusion boundary, leads to premature failure which propagates along the defects at or near the fusion plane. The novelty in this study lies in combining nanoindentation-informed finite element analysis (FEA) to evaluate the use of mechanical interlocking improve mechanical performance in the face of defective interfacial adhesion.

To assess the local mechanical properties across multi-material fusion boundary, nanoindentation is applied to determine the gradient of material hardness. Oliver and Pharr's analytical method to determine Young's modulus from nanoindentation is then used to inform FEA. The fusion boundary was modeled as an effective third material partition to generate a digital twin of the fusion boundary. By designing interlocking patterns, we aimed to alternate tensile and compressive stress across the metamaterial structure for an overall reduction in the stress concentration at the site of as-built cracks to reduce the defect's likelihood of propagating to the state of premature failure. Three mechanically interlocked designs are evaluated to demonstrate the capacity to locally load regions of the fusion boundary in compression despite global tensile loading.

Keywords: Finite Element; Interlocking Metamaterials; Nanoindentation; Multi-material AM; Multi-material LPBF

1. INTRODUCTION

Laser powder bed fusion (LPBF) is a rapidly developing laser-based metal additive manufacturing (AM) method which relies on a layer-by-layer deposition of metal powder following by the melting and solidification of a 3D object using a high-energy laser. This method is known for its ability to manufacture high-performance complex geometries with good geometric resolution and surface finish [1]. Recent technological advancements have allowed for selective powder deposition (SPD) which allows for a voxel-based approach to powder deposition, allowing for multiple metals to be deposited in a single layer. Figure 1 shows a schematic of the SPD process, where recoating drums establish a negative vacuum to maintain a layer of powder, with selective bursts of positive pressure at the bottom of each drum to release voxels of powder onto the powder bed. This process, called multi-material LPBF (MM-LPBF) sees two distinct regional allocations of powders, (1) the deposition of multiple materials across multiple layers and (2) the deposition of multiple materials within a single layer, allowing for 3D voxel-based functional gradation.

The application of MM-LPBF extends across a broad spectrum of industries, offering unpresidented geometrical resolution for multi-metallic designs due to its smaller laser spot size (50-80 μm) and thinner layer thickness (<100 μm) [2]. Multi-material AM has been previous demonstrated in directed energy deposition (DED). DED has a poorer geometric resolution (laser spot size ~0.25 – 2.0 mm, layer thickness 1.0 – 4.0 mm) than LPBF, and often requires subtractive post-processing (e.g. machining) to achieve near-net shape fabrication [2]. Notable applications of multi-

material AM include orthopeadic implants using tailored biocompatible mateirals [3].

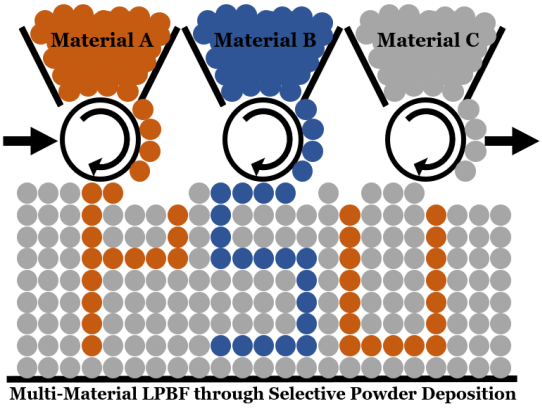


FIGURE 1: SCHEMATIC OF MM-LPBF ENABLED THROUGH SPD. RECOATING DRUMS ESTABLISH A NEGATIVE VACUUM MAINTAIN A COAT OF POWDER, SELECTIVE BURSTS OF POSITIVE PRESSURE AT THE BOTTOM OF EACH DRUM RELEASE ONE VOXEL OF POWDER ONTO THE POWDER BED TO ALLOW FOR THE SELECTIVE DEPOSITION OF POWDER IN 3D.

MM-LPBF enables additional applications, such as advanced heat exchangers, to be redesigned to include intricate, highly permeable triply periodic minimal surfaces (TPMS) to design and control fluid flow [4], [5]. Jointly, material assignment across the heat exchanger may allow for strength in load-bearing sites, and highly conductive material elsewhere. A multi-metallic redesign for a channel wall nozzle has been investigated for liquid rocket engines to reduce cost and schedules associated with fabrication by Grandl et al. Their proposed design includes a hot wall inner liner (C-18150 copper alloy) of a highly conductive material, and the closeout and outer jacket of Monel 400 nickel superalloy [4].

Wang et al. has reported a comprehensive review on the state of multi-material LPBF printed structure, which shows that most studies on MM-LPBF concentrate the microstructural characteristics and metallurgical bonding of the interface for a wide range of material pairings [6]. Table I provides a brief review of published literature which examines the metallurgical bonding of a bimaterial interface manufactured through MM-LPBF.

1.1. Prior Investigation of the Interfacial Region

Prior research has shown an interfacial region at the bimaterial fusion boundary, characterized by non-uniform chemical composition and grain structure [7]. Electron dispersive spectroscopy (EDS) has shown heterogeneous mixing of chemical composition at the fusion boundary which follows the visible contours of the laser scan path. Electron backscatter diffraction (EBSD) has shown extreme irregularity between grain structure (size, shape, orientation) of the fusion boundary, and grain structure typically associated with the constituent material for this process. According to EBSD, this region has been shown to extend to a maximum of ~250 μm as measured laterally across the fusion plane. Due to the unusual chemical composition, and the grain structure of this interfacial region, there is a need to investigate the transition of material properties across the fusion plane in MM-LPBF components. Optical microscopy has shown micron- and nano-scale cracking in the steel material section directly adjacent to the interface (Figure 2).

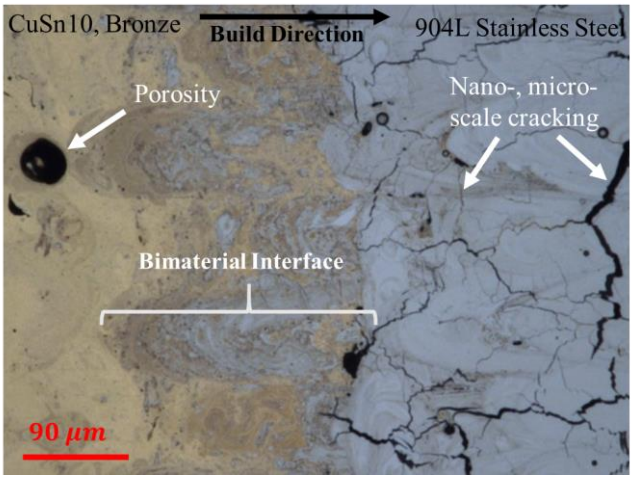


FIGURE 2: OPTICAL MICROSCOPY IMAGE OF THE BIMATERIAL INTERFACE BETWEEN BRONZE (LEFT) AND 904L STAINLESS STEEL (RIGHT) FABRICATED THROUGH MULTI-MATERIAL LPBF. THIS IMAGE WAS TAKEN FROM WITNESS CUBE (SS, BR)X (FIGURE 3 FOR ORIENTATION).

Thermomechanical simulation, combined with micro-CT scanning, has demonstrated that the cause of the cracking is due to process-induced thermal cycles, which occur on a layer-by-layer basis, combined with dissimilar thermal material properties [8]. The thermal-induced cracking phenomenon has been shown to be dependent on material orientation with respect to the build direction. The formation of intermetallics is a potential cause for crack exacerbation [9], [10], [11], [12], [13]. To explore this as a contributing factor for this alloy combination, energy dispersive spectroscopy (EDS) was performed to identify chemical compositional patterns which may be consistent with the formation of intermetallics.

Mechanical testing has shown these as-build defects (cracking, porosity) cause premature failure. Because these defects are concentrated around the fusion boundary, a novel design approach for the material interface may be beneficial in improving the overall mechanical performance of the bimaterial interface. Determining local properties of the fusion plane through nanoindentation enables finite element analysis (FEA) to be applied to evaluate the application of metamaterial design to mechanically interlock constituent material regions. The redesigned fusion boundary alternates tensile and compressive stress across the metamaterial surface for an overall reduction in the stress intensity concentration associated with as-build cracking with an aim to reduce the defect's likelihood of propagating to the state of premature failure.

TABLE I: A BRIEF REVIEW OF PUBLISHED SCIENTIFIC LITERATURE ON THE METALLURGICAL BONDING OF A MULTI-MATERIAL LPBF MANUFACTURED INTERFACE. PRESENTED IN ORDER ACCORDING TO PUBLICATION DATE.

Metal Multi-Material Interface Pairing	Notes on Interface metallurgical bonding
316L & C18400 [14]	Substantial Fe and Cu diffusion at the bond interface Good metallurgical bonding at the interface Highly refined microstructure at the interface attributed to rapid solidification.
AlSi10Mg & C18400 [9]	Good metallurgical bonding observed Intermetallic formation (Al_2Cu) at the interface due to elemental diffusion
Fe + Al-12Si [10]	Large amounts of cracking due to poor material compatibility and miscibility Interface showed high hardness, consistent with FeAl intermetallic
316L & CuSn10 316L & IN718 [15]	Good metallurgical bonding for both material pairs. Some defects (porosity, cracking) were found due to un-compressed powders during deposition
316L & IN718 [16]	Considerable cracking was observed on the 316LSS side of the interface. Defects at the interface are attributed to the mismatch between laser energy density and material properties.
316L & CuSn10 [17]	Good metallurgical bonding, some dendritic cracking near the interface, within the 316LSS region. Cracking perpendicular to the interface impacted mechanical performance. Defects attributed to a difference in thermal expansion/conduction between materials. Mechanical performance of the interface was below that of either constituent material.
CuSn10 & 18Ni300 [18]	Microhardness at the interface exhibits a gradient distribution between constituent materials Remelting (twice, and three times) did not increase bonding strength
Pure Al and Fe [11]	Formation of intermetallics (Fe_2Al_5) causing 'flaking' between layers Interface bond quality impacted by process parameters (laser power, speed, hatch distance)
316L & C52400 Cu [12]	No Fe-Cu intermetallic compounds were found at the interface Increasing the degree at which the metals mix within the melt pool produced excellent metallurgical bonding, and reduced the formation of defects (cracking, oxides, pores). Defects attributed to dissimilar thermal expansion, heat conduction and melting points. Interfaces display finer grains, on average, than otherwise expected from single material fabrication.
Ti6Al4V & In718 [13]	Defect-free grading was achieved from 0-20 wt.% of IN718 Severe cracking at the interface attributed to Ti_2Ni phases occurring at the liquid state Microhardness shows gradient distribution between constituent materials.
316L & CuSn10 [19]	Re-melting and recrystallization refinement improved mechanical strength of interface Cracking at the interface attributed to difference in thermal properties and copper penetration
CuSn10 & Ti6Al4V [20]	Marangoni convection-induced circular flow within the melt pool is a driving force for elemental diffusion Melt pool size (especially depth) is linked to the diffusion of elements between alloys, and the bonding strength of the interface. Remelting near the interface increased the hardness of the interface. However, keyhole pores were found post-remelt.
316L & CuSn10 [21]	Cracking and porosity observed at the interface High volumetric energy density, low scanning speed (to allow more elemental diffusion), and selective build strategy (diffusion of elements based on material density and gravitational pull) are all conductive to strengthening the metallurgical bonding.
Ti & Ta IN718 & GRCop-42 [22]	Thermomechanics, defect behavior, and microstructure formation (thermal history and melt pool dynamics) are dependent on material orientation.
904L & CuSn10 [8]	Determines interfacial bonding is, in part, dependent on material orientation with respect to the build direction. Observed cracking at the interface occurred within the steel and is attributed to thermomechanical difference between materials.

The focal point of the experimental characterization in this study is to (1) determine contributing causal sources for cracking near the interface, and (2) explore the use of nondestructive evaluation to determine the gradient of material properties across the bimaterial interface. This study will include the use of EDS to determine the presence of intermetallics, and their contribution to as-build cracking, and nanoindentation across the bimaterial interface to determine local material properties. Using these data, FEA will be used to evaluate the proposed redesign of the bimaterial interface. Provided the direction and location of potential cracks near the fusion plane, increased stress concentrations in these areas significantly increase the stress intensity factors across all cracks which run parallel to the fusion plane. As a result, the likelihood of these cracks propagating through the entire cross section of the interface is considerably high. The goal in redesigning the fusion boundary is to interrupt the continuation of stress across the interface to prevent cracks from propagating across the entire boundary, and potentially causing premature failure.

2. METHODOLOGY

2.1 Witness Cube Design and Fabrication

1 cm^3 witness cubes were printed in three build directions with respect to the material fusion plane, shown in Figure 3.

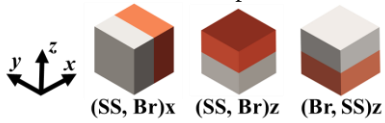


FIGURE 3: WITNESS CUBE BUILD ORIENTATION, WHERE THE NOMENCLATURE INDICATES THE SEQUENTIAL MATERIAL PLACEMENT ALONG THE CARTESIAN DIRECTION NORMAL TO THE FUSION PLANE.

The witness cubes were fabricated on an Aconity MIDI+ (Aconity3D GmbH, Germany) equipped with an Aerosint SPD v1 (Aerosint SA, Belgium), with a single mode fiber laser at a continuous wavelength of 1070 nm and spot size of 80 µm. The build chamber was pumped with argon gas to maintain O_2 concentrations below 500 ppm. Material regions were sintered according to the laser processing parameters detailed in table II.

TABLE II: WITNESS CUBE MM-LPBF PROCESSING PARAMETERS BY MATERIAL

Material	Power (W)	Scanning Speed (mm/s)	Hatch Spacing (µm)	Layer Thickness (µm)
Bronze	200	500	120	40
904L SS	250	750	80	40

2.2 Investigation of Intermetallics

EDS (Apreo 5) analysis was performed to provide additional information regarding the chemical composition and distribution of elements across the interface. Microscopy was performed at an operating voltage of 20kV and 3.2mA at a working distance of 10mm Cube (Br, SS)z was examined using several EDS line scans (shown in Figure 8) across the cracks found around the fusion plane. This cube served as a representative specimen for micron- and nano-scale cracking observations.

2.3 Design of Interlocking Metamaterials

In solid mechanics, stress concentration is taken as regions within an object which experiences significantly greater stress than the surrounding regions. Stress concentrations tend to occur when there are irregularities in the geometry or material of a load-bearing component which cause an interruption of stress distribution. Due to the abrupt material change, the bimaterial region is taken to be a stress concentration. Not only is there a severe change in mechanical properties across the fusion plane, but the interface itself is seen as having irregular composition and grain structure. In the case of a flat fusion boundary, as seen in the witness cube, the concentration of stress is highly focused at the interface, extending across the fusion plane.

Previous work in multi-material extrusion AM has investigated this same concept to increase the interface between two polymers.

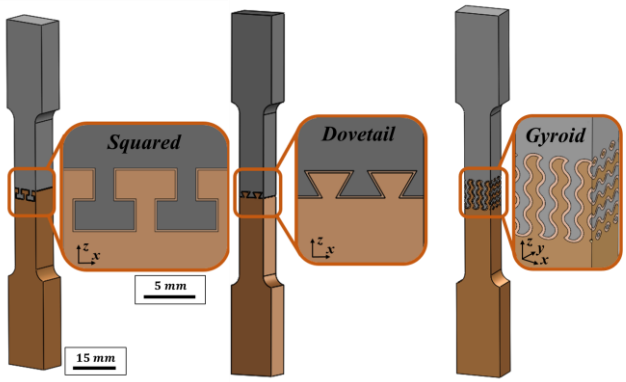


FIGURE 4: IMAGE OF MULTI-MATERIAL TENSILE MODELS WITH REDESIGNED BIMATERIAL INTERFACES. MATERIAL ASSIGNMENTS PLACE STAINLESS STEEL ON THE TOP, AND BRONZE ON THE BOTTOM.

The three designs (Figure 4) are inspired by previously studied bimaterial interfaces or non-adhesive joining methods with the goal of strengthening their connection [23], [24], [25].

Design 1: Squared fusion boundary, characterized by periodic, T shaped interlocking regions. This boundary geometry places portioned of the interface in full compression, or shear.

Design 2: Dovetail fusion boundary, characterized by periodic, trapezoidal interlocking regions. This boundary geometry should also result in the slanted bimaterial regions experiencing shear / compressive stresses.

Design 3: Interlocking gyroid fusion boundary, modeling using two interlocked solid gyroid regions made of bronze and 904L SS, and an interface modeling using a shell gyroid.

2.4 Nanoindentation

Prior investigation has found thermomechanical, processes-induced cracking in the steel material section adjacent to the fusion plane in witness cubes (SS, Br)x and (Br, SS)z **[8]**. Witness cube (SS, Br)z demonstrated good metallurgical bonding at the interface and minimal visible defects at the surface. To avoid unintentionally indenting into a crack, witness cube (SS, Br)z was selected for nanoindentation.

Young's modulus and hardness were measured with a nanoindenter (Hysitron TI 950, Minneapolis, MN) fitted with a Berkovich tip. Results from a total of 21 indents were obtained at 7 locations across the fusion plane ($n = 3$ for each location, Figure 5). Two locations were indented on the far side of either material for as-built bronze and 904L stainless steel properties. Seven locations were taken in a line perpendicular to the fusion plane with an approximate breadth spanning 500 $\boldsymbol{\mu m}$. This line was repeated in three replicates, and the results for each location were averaged. Indentation was conducted at a maximum cutoff load of 6 mN and a dwell time of 2 seconds. Sample number 3.1 was removed from the data set as it appeared to have indented into a crack or deformity. This was observed by a long displacement region with little to no change in applied load.

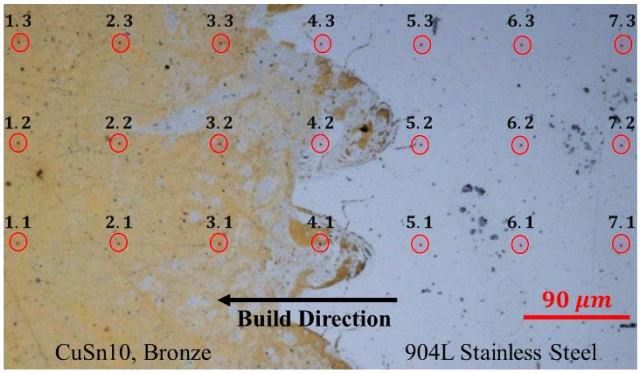


FIGURE 5: OPTICAL MICROSCOPY IMAGE OF THE NANOINDENTATION LOCATION ACROSS THE BIMATERIAL INTERFACE. A 7X3 GRID WAS ESTABLISHED TO EVALUATE THE GRADIENT OF MATERIAL PROPERTIES.

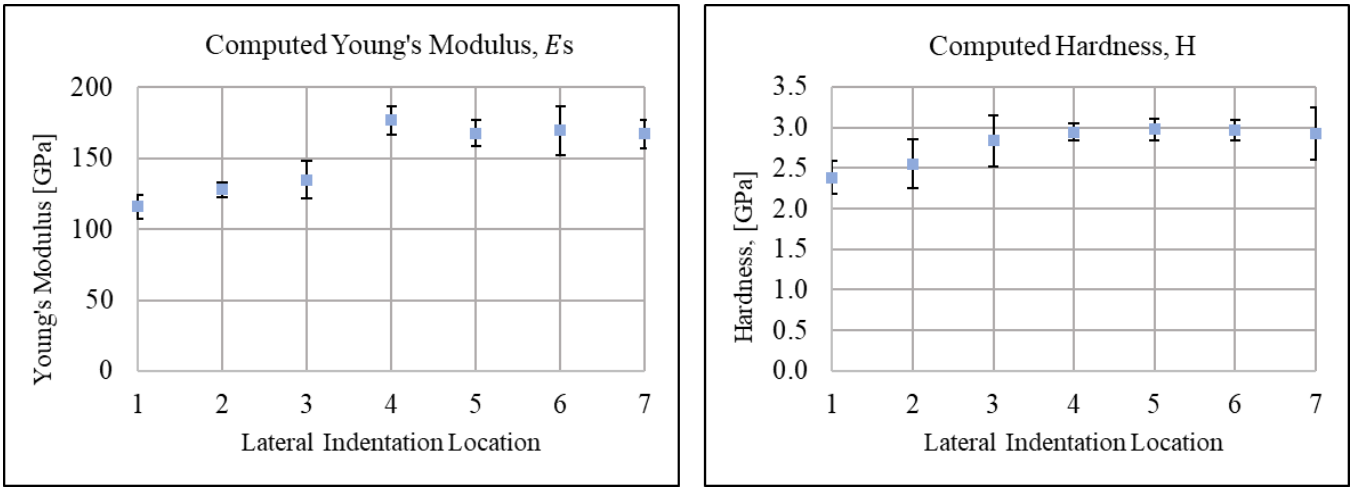


FIGURE 6: COMPUTED YOUNG'S MODULUS AND HARDNESS [GPA] FOR THE SAMPLE BASED ON THE OLIVER AND PHARR NANOINDENTATION METHOD. MOVING LATERALLY ACROSS THE FUSION BOUNDARY, FROM BRONZE TO STEEL (FIGURE 5 FOR LATERAL LOCATION OF INDENTATIONS). REPORTED ERROR IS ± 2 STANDARD DEVISIONS FROM THE MEAN.

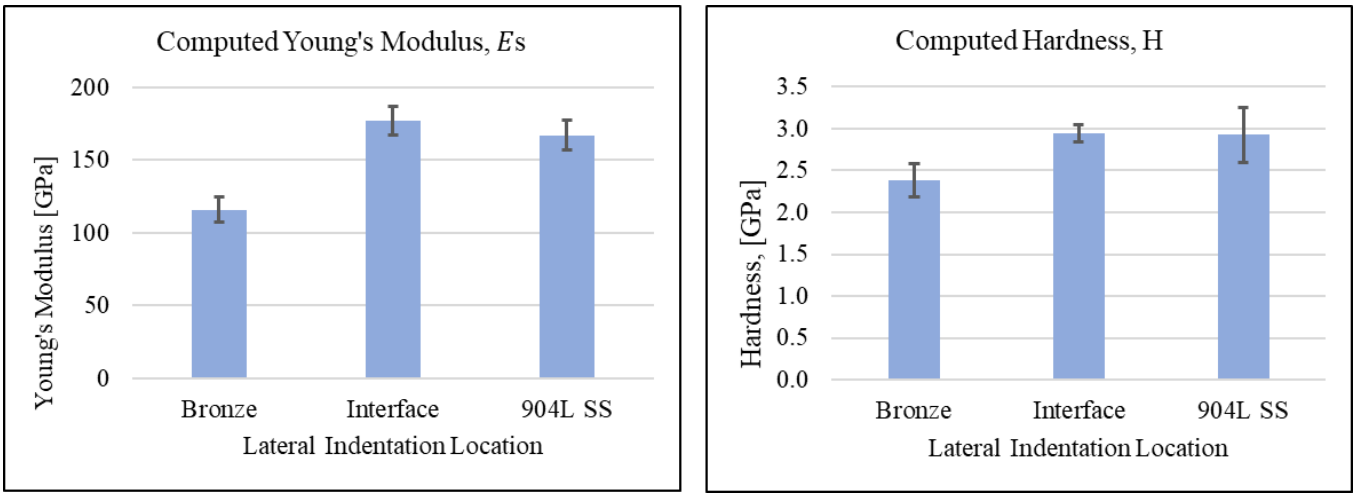


FIGURE 7: COMPUTED YOUNGS MODULUS AND HARDNESS [GPA] BASED ON THE OLIVER AND PHARR METHOD, COMPARING VALUES FOR EACH CONSTITUENT MATERIAL SECTION VERSUS THE INTERFACIAL REGION.

The TriboScan (10.2.0.1) software was used during scanning to automatically detect the slope of the unloading cure to produce the reduced Young's modulus (E_r). The sample Young's modulus was then calculated based on the Oliver and Pharr method outlined in Appendix A. For this calculation, the Poisson's ratio of all materials must be assumed. In this case, the Poisson's ratio of this bronze alloy was taken as 0.34 [26], and the Poisson's ratio of 904L SS was taken as 0.33 [27]. Since the Poisson's ratio of the fusion boundary cannot be estimated using prior literature, it was assumed to be 0.30, as is the average value for most metallic materials [28]. Indenter locations 1.1, 1.2 and 1.3 are in regions of anticipated microstructure of bronze, and locations 7.1, 7.2 and 7.3 are consistent with microstructure expected of 904L SS. As a result, these locations should be far enough from the interface to be representative of the solid material regions.

2.5 Finite Element Modeling and Analysis

Numerical analysis was performed using Ansys 2023 static simulation. Each material section was modeled as parts and connected using bonded contact. Material sections were meshed independently using tetrahedral element shapes. The bottom surface of the tensile model was fixed in all directions, and a 5kN external load was applied to the opposite surface. Element sizes were restricted to the range of $\mathbf{0.125}\ \boldsymbol{mm}$ and $\mathbf{0.25}\ \boldsymbol{mm}$.

Nanoindentation has shown a non-discrete boundary of mechanical properties across the fusion zone. To incorporate the mechanical and material behavior of the fusion boundary, it was modeled as an effective third 'digital' material. This third material was modeled as a $250\ \mu m$ thick surface between the two solid material regions.

3. Results

3.1 Investigation of Intermetallic Formation

This study included EDS to elucidate potential correlations between material composition and cracking phenomena. EDS and EBSD scans were conducted across the cracked as-built surface of (Br, SS)z (steel sintered over bronze) witness cubes. Map scans were performed across the interface (Figure 8B, 8D, 8E), and line scans (Figure 8C, 8F) were taken at

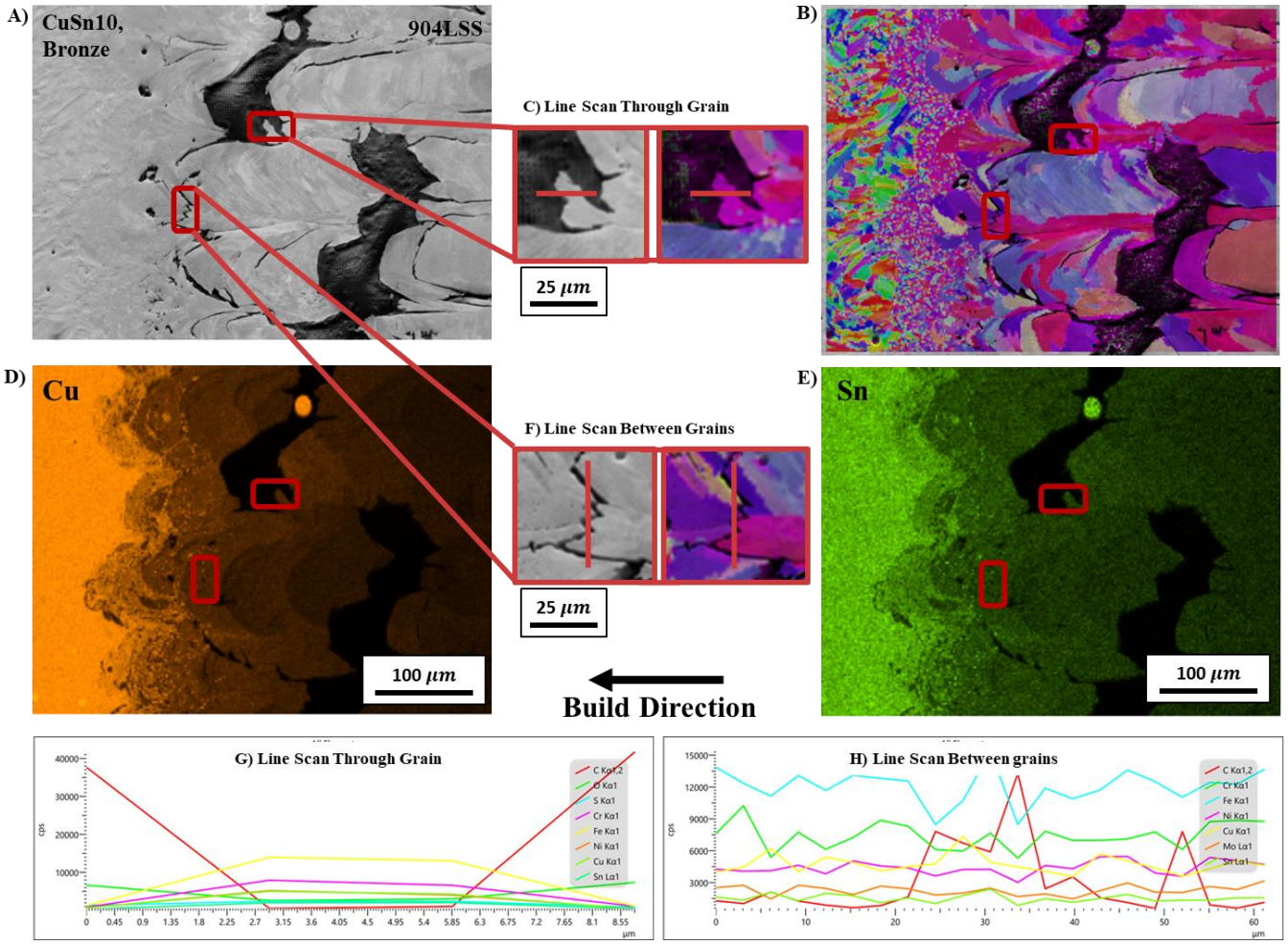


Figure 8: EBSD and EDS results from an investigation of intermetallic formation for specimen (Br, SS)z. A) SEM image of orientation witness cube fusion zone, B) EBSD grain boundaries for fusion zone with cracking, C) Line scan 1 near the edge of major cracking, D) EBSD element map of trace copper contamination through fusion zone and adjacent steel, E) EBSD element map of trace tin contamination through fusion zone and adjacent steel, F) Line scan 2 near the edge of minor cracking, G) Line scan 1 results by element weight percentage, H) Line scan 2 results by element weight percentage.

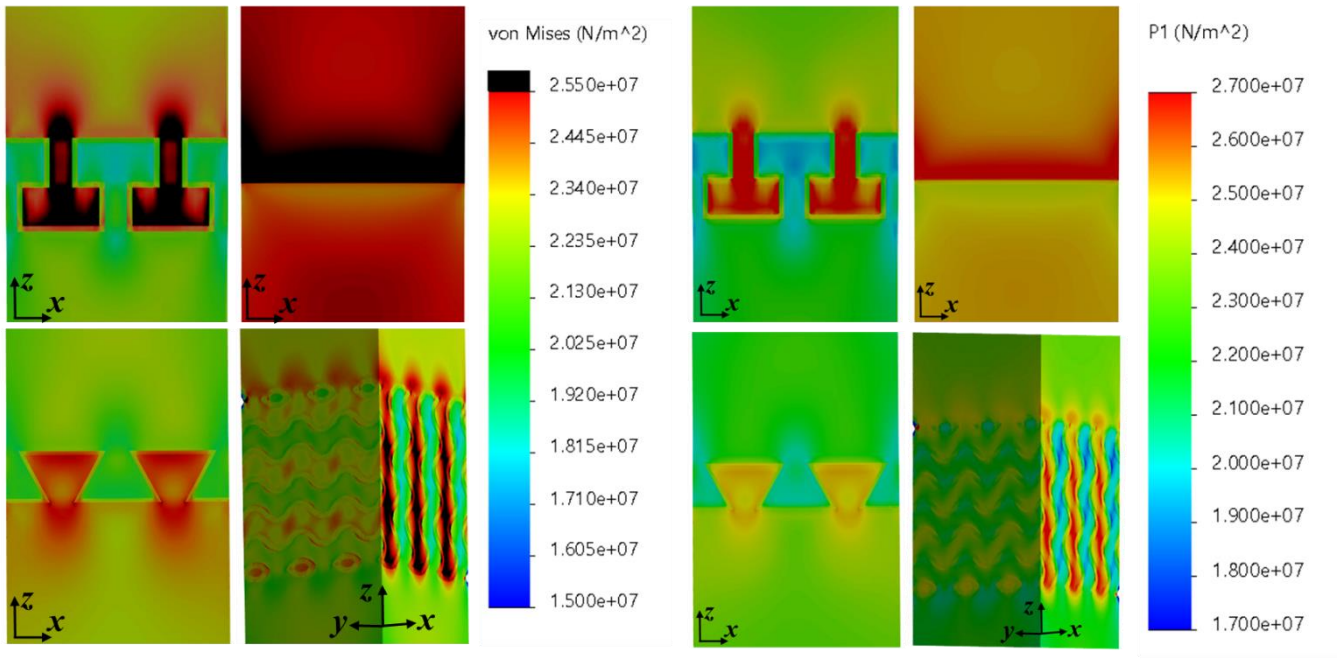


FIGURE 9: A) FEA RESULTS FOR THE DISTRIBUTION OF VON MISES STRESS ACROSS THE FUSION BOUNDARY. BLACK REGIONS DENOTE STRESS ABOVE THE IDENTIFIED THRESHOLD OF 25.5 MPA. THE UPPER MATERIAL IS 904L SS, AND THE BOTTOM IS BRONZE. B) FIRST PRINCIPAL STRESS DISTRIBUTION. THE UPPER MATERIAL IS 904L SS, AND THE BOTTOM IS BRONZE.

cracks which occurred through and across grain boundaries. EDS map scans were conducted across the interface to allow for the visualization of the spatial distribution of different element, which can help identify any elemental variations or anomalies associated with the cracking phenomena. The chemical (wt.%) composition across any line scans obtained do not reveal any discernable intermetallic phase associated with the cracking. Intermetallic phases are often implicated in the initiation and propagation of cracks in the joining of dissimilar metals, but their absence in this study suggests that other factors may be influencing the cracking behavior.

3.2 Nanoindentation for Young's Modulus

Reduced Young's modulus (E_r) and indentation hardness (H) were calculated by the TriboScan software, based on the model of Oliver and Pharr (Appendix A). Table III summarizes the average quantitative results for both mechanical properties at each lateral location across the fusion boundary.

TABLE III: INDENTER YOUNG'S MODULUS AND HARDNESS, BASED ON THE OLIVER AND PHARR METHOD.

Location	E_s [GPa]	H [GPa]
1	115.95	2.38
2	127.83	2.55
3	134.92	2.83
4	176.69	2.94
5	167.64	2.97
6	169.56	2.97
7	167.09	2.92

There is a notable spike in Young's modulus at location 4 (Figure 6, 7). This location is situated within a laser scan path of the first layer of copper sintered on top of the steel half of the witness cube. Prior EDS and EBSD has suggested that this region is characterized by non-uniform grain structure and mixed chemical composition. The increased Young's modulus is anticipated to raise the stress within the fusion boundary.

A review of the literature shows expected hardness of 904L stainless steel to be in the range of 2 – 3 GPa. The results from indentation location 7 (steel side of the fusion boundary) show a hardness of 2.92 GPa, which is consistent with this expectation [29]. Likewise, the expected hardness of CuSn10 bronze alloy to be in the range of 2 - 2.5 GPa [21], [30], which is similar to the hardness determined at location 1.

3.3 FEA *for* Interfacial Design Performance

The proposed redesigned interfaces are compared to the standard design for testing interface bonding strength for tensile specimens manufactured through MM-LPBF [17], [18], [19]. Results from FEA are shown in Figure 9. To directly compare the distribution of stress through the FEA results, a threshold of 25.5 MPa was set as a criterion for identifying regions of high stress concentration along the fusion boundary. This threshold allows for a standardized comparison across the different designs.

4. DISCUSSION

4.1 Investigation of Intermetallic formation

One of the primary objectives of this study was to assess whether the observed cracking in the MM-LPBF parts could be attributed to the formation of intermetallic compounds, particularly involving copper and tin in conjunction with iron. This study focused on cracks that formed both along and

across grain boundaries. Contrary to expectations, the results in Figure 8 reveal no evidence of intermetallic compounds at the edges of the observed cracks.

This outcome provides significant insight into the underlying mechanisms of cracking in multi-material LPBF components. The absence of intermetallic compounds along the cracks suggests that other factors, such as microstructural defects, residual stresses, or thermal gradients, or differences in cooling rates, may play more prominent roles in the crack initiation and propagation processes. This finding challenges the prevailing notion that intermetallic formation is a primary driver of cracking in MM-LPBF. This notion is further supported by the good metallurgical bonding observed in witness cube (SS, Br)z, in comparison to the defects found in the other two orientations.

4.2 Interfacial Site-specific Properties

Hardness increases across the fusion boundary, with a notable maximum hardness of 2.92 GPa reported at the center of the interface (location 4). Liu et al. reported a similar phenomenon occurring during the joint sintering of 316L stainless steel and CuSn10, where the hardness of the interface was measured as over 18% greater than that of the steel. Other publications have reported a linear gradient of hardness across bimaterial interfaces for CuSn10 & 18Ni300 and Ti6Al4V & In718 material pairs [13], [18]. The increased hardness at the interface may be attributed to an increase in grain size refinement along the interface, as apparent by preliminary EBSD results. These finer grains impede dislocation, and are associated with an increase in hardness. Further work using electron backscatter diffraction (EBSD) can quantify the relative degree of grain refinement. In addition, strain hardening may impact interfacial hardness as a result of the material's chemical and physical immiscibility at the interface [31]. Tan et al. cites the potential for this to induce micro-strain and distortion at the bonded region and report an overall improvement in interface strength as a result [31].

The curve of the computed Young's modulus showed a similar trend, with 904L stainless steel and CuSn10 measuring as 167.09 GPa and 115.95 GPa, respectively. At the interface, the Young's modulus increased to a peak of 176.69 GPa. Bulk 904L properties are cited to be closer to 200 GPa [27]. A lower Young's modulus could indicate an increase in expected ductility for this material. Bulk CuSn10 is known to have an expected Young's modulus of around 110 GPa, which is within 6% of the results found through nanoindentation [32].

4.3 Performance Evaluation of Redesigned Fusion Boundary

The redesign of a bimaterial interface is shown to alleviate the distribution of stress concentration as compared to a conventional flat interface. By incorporating interlocking material regions, the designs are able to redistribute stress into the solid materials regions, and away from the defective fusion boundary. In this way, the fusion boundary is seen to alternate between regions of high and low stress, as opposed to have uniformly high stress focused strictly at the fusion plane.

In the event that the interface experiences as-build cracking near the fusion plane, we anticipate the use of interlocking material regions will disrupt the anticipated path of crack propagation along the fusion plane. The presence of low-stress regions acts as a barrier, preventing crack propagation beyond regions of high stress, mitigating the risk of catastrophic failure [33]. Seen in figure 9, the stress distribution at the surface of the flat boundary is consistent, and above the threshold across the entirety of the specimen. From this, it is anticipated that this specimen will reach failure prematurely when compared to the redesigns.

According to the threshold stress, which is consistent across 100% of the conventional flat interface, the square design only experiences threshold stress across approximately 55% of its interface. Most of this stress is concentrated within the steel regions at the bottom of the interface. This is noteworthy, as the interface in contact with the surface is typically the site of crack propagation, however, this region does not exhibit a relatively high stress in the redesigned square part. Similar research has indicated a trend for fracture to originate at the interface surface, due to weak interfacial bond strength between steel and bronze alloys [21]. This suggests that, for the squared interface design, crack propagation will likely initiate from the center of the gage section outwards.

Similarly, the gyroid specimen maintains the highest stress concentration strictly within the steel regions. However, this design may benefit from interlocking material regions in to three dimensions using a minimal surface geometry with zero mean curvature. The lack of nodal struts, characteristic of TPMS surfaces, limits local regions of high-stress concentration. This has been shown in literature through the mechanical comparison between strut-based lattices and TPMS surfaces [34], [35], [36].

The dovetail design appears the most mechanically resilient with the lowest portion of the interface at the threshold stress, (only the inner corners of the design reaching the threshold). In this design, the sloped regions see a decrease in Von Mises stress, as indicated by regionally negative 3rd principal stress. This, in part, demonstrates that these regions see compressive stress in at least one of the primary directions.

Immediate future research efforts will include the fabrication and mechanical testing of all four tensile specimens. From the evident FEA results here, it is anticipated that the conventional flat interface design will underperform as compared to the three alternative design choices.

5. Conclusion

The interface characteristics of 904L stainless steel/CuSn10 Bronze cubes manufactured through MM-LPBF have been investigated. Local material characteristics of the interface were used to inform FEA to determine the effectiveness of

the redesign of the bimaterial interface at redistributing stress. The key results are as follows:

1. Intermetallics were not determined to be a contributing factor towards the formation of cracking near the interface. It is suspected that cracks are formed due to process-induced residual stresses.
2. The strength of metallurgical bonding was found to be partially determined by material orientation along the build direction. Good metallurgical bonding was found for the interface produced by sintering bronze over a 904L SS substrate. Severe cracking was found for interfaces produced by sintering 904L SS over bronze.
3. The interfacial region's hardness was higher than that of either constituent material. This may be attributed to the refinement of grains within the interface.
4. Redesigning the interfacial region to include interlocking feature is a promising method to reduce stresses which may reduce the likelihood of micro- and nano-scale crack propagation across the interface under loading.

Ongoing research, guided by the innovative design application of MM-LPBF and ICME (Integrated Computational Materials Engineering) computational tools, is driving us closer towards realizing the full application of high-resolution MM-AM and its transformative impact on AM design. Immediate future research efforts will expand on this work to include experimental validation of the presented FEA by comparing simulation to experimental strain fields found through digital image correlation (DIC).

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Appendix A: The Oliver and Pharr method of analytically determining Young's modulus from nanoindentation

NOMENCLATURE

H	Hardness [GPa]
P	Applied Load [N]
P_{max}	Maximum Applied Load [N]
A_r	Residual indentation area [mm^2]
E_r	Reduced Young's modulus [GPa]
E_s, v_s	Youngs Modulus, Poisson's ratio of sample
E_i, v_i	Youngs Modulus, Poisson's ratio of indenter
h	Penetration depth [mm]

Nanoindentation is a nondestructive material testing technique used to measure mechanical properties at the surface non-destructively. It is constituted of a controlled load application to a sharp indenter top (typically diamond material), and measuring the force applied against the depth of indenter penetration [37]. Material hardness, H, is then defined by the maximum applied load, P_{max}, and the residual indentation area, A_r.

$$H = \frac{P_{max}}{A_r} \tag{1}$$

Where the residual indentation area is found as the indenters contact area projected onto the normal loading plane. Oliver and Pharr have proposed a widely adopted method for analytically determining the reduced Young's Modulus using the un/loading-displacement curve and indentation projected area [38].

$$E_r = \frac{\sqrt{\pi}}{2\beta}\frac{\frac{dP}{dh}}{\sqrt{A(h_{cm})}} \tag{2}$$

Where $\frac{dP}{dh}$ is found as the initial slope of the unloading-displacement curve, shown in Figure A1, and β is a constant determined by the indenter's size.

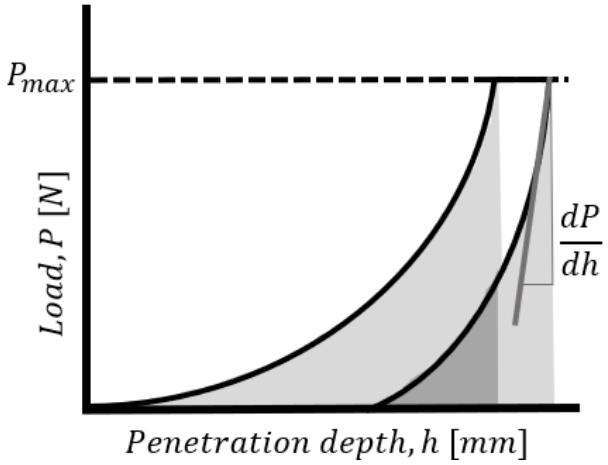


FIGURE A1: SCHEMATIC OF LOAD-DISPLACEMENT LOADING AND UNLOADING CURVE FOR AN INSTRUMENTED NANOINDENTATION TEST.

The analytical Young's modulus of the sample is then found as Equation 4 according to the Oliver and Pharr model. Where the Youngs modulus and Poisson's ratio of a diamond indenter is known to be 1141 MPa and 0.07 respectively [39].

$$\frac{1}{E_r} = \frac{1 - v_s^2}{E_s} + \frac{(1 - v_i^2)}{E_i} \tag{3}$$

$$E_s = \frac{(1 - v_s^2)}{\frac{1}{E_r} - \frac{1 - v_i^2}{E_i}} \tag{4}$$

Thus, Equation 2 may be used to empirically determine E_r through load-displacement data and analysis, and substituted into Equation 4 to calculate the sample's Young's modulus as per the Oliver and Pharr method. Wherein the only assumption necessary in the context of this study is the assumed Poisson's ratio of the sample materials.